

# Image Fragment Reconstruction via Adjacency Prediction

## Case Study Report

### 1. Model Architecture

The model is a **Siamese CNN** that operates on pairs of  $16 \times 16$  fragments. A shared encoder maps each fragment to a 256-dimensional embedding; a single comparison head maps the two embeddings to a scalar confidence score  $p_{ij} \in [0, 1]$ .

**Encoder.** Three Conv–BatchNorm–ReLU blocks with max-pooling after the last two reduce the spatial dimensions from  $16 \times 16$  to  $4 \times 4$ ; a linear layer projects the flattened features to 256-d (ReLU + dropout  $p = 0.3$ ).

**Comparison head.** Inspired by InferSent [1], the head takes the 1024-dimensional vector  $[\mathbf{a} - \mathbf{b} \parallel \mathbf{a} \odot \mathbf{b} \parallel \mathbf{a} \parallel \mathbf{b}]$  as input and produces a scalar confidence score via one hidden layer (ReLU).

**Why Siamese?** Weight sharing ensures the comparison is symmetric and forces the encoder to learn a universal fragment representation rather than one tied to a specific input slot. No image-level labels are needed: the only supervision is whether two fragments are adjacent, which follows directly from their grid positions.

**Why adjacency prediction?** We preferred this over contrastive approaches because the adjacency signal is local and geometrically interpretable — each label has a direct spatial meaning — which makes failure modes transparent and mirrors the edge-continuity cues humans use when solving jigsaw puzzles. This made it easier to reason about bottlenecks and diagnose where the model struggled.

### 2. Training Strategy

**Pretext task.** The model is trained to predict spatial adjacency:  $y_{ij} = 1$  iff fragments  $i$  and  $j$  are grid-neighbours within the same image. With  $N = 10$  images on a  $4 \times 4$  grid, the negative-to-positive ratio is exactly  $12,480/240 = 52$ . We use weighted BCE with tilt parameter  $\beta$ :

$$\mathcal{L}_{\text{WBCE}}(\beta) = -\frac{1}{N} \sum_{(i,j)} [\beta \cdot 52 y_{ij} \log p_{ij} + (1 - y_{ij}) \log(1 - p_{ij})].$$

**Multi-task loss.** Adjacency is a local signal; the downstream task is global clustering. A same-image loss term

$\mathcal{L}_{\text{BCE}}^{\text{same}}$  is added to bridge this gap, acting on the same score  $p_{ij}$ :

$$\mathcal{L} = \lambda_{\text{adj}} \mathcal{L}_{\text{WBCE}}(\beta) + \lambda_{\text{same}} \mathcal{L}_{\text{BCE}}^{\text{same}}.$$

**Inference.** To assign each fragment to its source image, the pairwise scores must be turned into hard cluster assignments. The  $160 \times 160$  matrix of scores  $p_{ij}$  is used directly as a similarity matrix for **balanced spectral clustering**: spectral embedding finds clusters as well-connected subgraphs of the similarity graph, while the Hungarian-assignment balanced  $k$ -means step enforces exactly 16 fragments per cluster, respecting the fact that every source image contributes the same number of patches. Each inference call processes one batch of 10 images ( $N = 160$  fragments):  $N$  encoder passes followed by  $\binom{N}{2} = 12,720$  head evaluations ( $\mathcal{O}(N^2)$ ) and an  $\mathcal{O}(N^3)$  spectral decomposition. At this fixed scale both steps complete in under a second.

**Training details.** 1.28M training images (ImageNet-64); each step samples 10 fresh images. Up to 25,000 iterations; early stopping with patience 10, evaluated every 250 steps over 20 validation batches. All runs on CPU (Apple M-series).

### 3. Evaluation Results

Performance is measured by the **Adjusted Rand Index** (ARI), which counts pair-level agreement between predicted and true clusters, adjusted for chance.  $\text{ARI} = 0$  is random;  $\text{ARI} = 1$  is perfect.

**Validation results.** A five-seed study (seeds 0–4) across five hyperparameter configurations (Table 1) shows that conditions (a)–(d) cluster tightly in the range 0.53–0.56 ( $\sigma \approx 0.01$ –0.02), while condition (e) ( $\beta = 1$ , class-balanced weighting) sits noticeably lower at 0.46. A Welch one-way ANOVA finds a statistically significant difference ( $\alpha = 0.10$ ), driven by condition (e); the loss weights within conditions (a)–(d) are not a strong lever at this scale.

**Test results.** The best checkpoint (run `multitask_10_seed_2`) was evaluated on 1,000 independent test batches:  $\text{ARI} = 0.521 \pm 0.104$ ,  $\text{NMI} = 0.686 \pm 0.071$ ,  $\text{purity} = 0.694 \pm 0.076$ ,  $\text{adjacency F1} = 0.356$ ,  $\text{AUROC} = 0.948$ . In concrete terms, a purity of 0.694 means that on average approximately 5 out of 16 fragments per cluster are assigned to the wrong source image.

| Configuration                    | Mean ARI | SD    |
|----------------------------------|----------|-------|
| (a) adj-only, $\beta = 1/52$     | 0.538    | 0.016 |
| (b) same-only                    | 0.541    | 0.018 |
| (c) multi-task $\lambda_s = 1.0$ | 0.557    | 0.009 |
| (d) multi-task $\lambda_s = 0.2$ | 0.531    | 0.012 |
| (e) adj-only, $\beta = 1$        | 0.460    | 0.019 |

Table 1: Validation ARI (5 seeds each).

**Pretext task.** Adjacency AUROC sits at 0.95–0.97 across all configurations, indicating the model has nearly solved the binary pretext task. The remaining gap to perfect clustering is consistent with the fragment-ambiguity cases discussed in Section 4.

## 4. Challenges and Improvements

**Class imbalance.** The 52:1 negative-to-positive ratio in adjacency pairs requires careful loss weighting. The multi-seed study shows that the textbook class-balanced weight ( $\beta = 1$ ) degrades ARI by  $\sim 0.08$  relative to plain BCE ( $\beta = 1/52$ ), suggesting that aggressive upweighting forces the model to chase ambiguous positive pairs at the cost of overall clustering quality.

**Local vs. global signal.** Adjacency is a nearest-neighbour signal; two fragments from opposite corners of the same image are labelled negative. The same-image loss term partially closes this gap but does not produce a statistically significant improvement at this scale, suggesting the binding constraint is fragment ambiguity rather than the loss formulation.

**Fragment ambiguity.** The failure-modes analysis shows that near-uniform or repetitive-texture patches produce similar embeddings regardless of source image, and are frequently mis-assigned.

## 5. Next Steps

**Pretrained encoder.** Replacing the trained-from-scratch CNN with a self-supervised encoder (e.g. DINOv2) would provide richer features at  $16 \times 16$  resolution at no labelling cost, likely the highest-impact single change.

**Topology-aware clustering.** The true clustering is more constrained than “equal sizes”: each cluster’s induced subgraph must be isomorphic to a  $4 \times 4$  grid. A GNN-based approach that exploits this planarity constraint could resolve borderline assignments that spectral clustering misses.

**Larger-scale statistical study.** With  $n = 5$  seeds the ANOVA is underpowered for effect sizes below  $\sim 0.07$  ARI. A 20-seed study would give  $\Delta_{\min} \approx 0.03$ , sufficient to detect the differences suggested by the single-seed sweep.

**Color-histogram baseline.** A non-learned baseline using per-fragment color histograms with spectral clustering would anchor the model’s ARI against a trivial reference and quantify how much benefit comes from learned representation versus color matching.

## References

- [1] Conneau et al. (2017). *Supervised Learning of Universal Sentence Representations from Natural Language Inference Data*. EMNLP.

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